

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Problem-Solution Fit

Problem:

- Manual data import from external sources is time-consuming
- Difficult to maintain relationships between User, Department and Location
- High risk of data inconsistency and duplication
- Complex data mapping across multiple tables

Solution:

- Use **Import Sets** to automate data ingestion from external sources (CSV, etc.)
- Apply **Transform Maps** for structured and accurate field mapping
- Leverage **Dot Walking** to easily establish relationships between tables
- Ensure consistent, scalable, and error-free data integration



Proposed Solution Template

Date	15 February 2025
Team ID	SIATID-2026-3651
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	2 Marks

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Organizations face challenges in importing large volumes of external data manually, which leads to errors, duplication, and inconsistent records. Additionally, maintaining relationships between related tables (like User, Department, and Location) is complex and time-consuming without automation.
2.	Idea / Solution Description	The solution uses Import Sets to load external data (CSV, Excel, etc.) into the system and Transform Maps to map source data to target tables automatically. Using Coalesce fields, duplicate records are avoided. ☒ Dot Walking is applied to establish and populate relationships, between tables (e.g., User → Department → Location), ensuring accurate data linking and automation.
3.	Novelty / Uniqueness	This solution uniquely combines automated data import with intelligent relationship mapping. The use of Dot Walking simplifies handling complex relational data without writing extensive scripts, making the process efficient and user-friendly, compared to traditional manual mapping methods.
4.	Social Impact / Customer Satisfaction	
5.	Business Model (Revenue Model)	Provides scalable handling of large datasets through automation.
6.	Scalability of the Solution	Highly scalable for handling increasing volumes of data and expanding relational complexity across multiple tables.

Automating Data Import and Relationship Mapping

Using Import Sets & Dot Walking

A ServiceNow-based solution to seamlessly import, transform, and map data into related records.

Key Components

- **Data Source** (CSV, API, Database)
- **Import Set Table** (Staging Area)
- **Transform Map**
 - ✓ Field Mapping
 - ✓ Coalesce Logic
- **Target Table**
 - ✓ User, Incident, CMDB)
- **Dot Walking**
 - ✓ Cross Table Referencing)
- **Business Rules & Scripts**
 - ✓ Validation & Automation)

Architecture Flow



How It Works



Outcome

- ✓ Automated Data Import
- ✓ Accurate Relationship Mapping
- ✓ Reduced Manual Effort
- ✓ Improved Data Consistency